MIL-P-551E
21 September 1983
SUPERSEDING
MIL-P-551D
29 September 1972

MILITARY SPECIFICATION

PIPES AND ELBOWS, AIR CONDITIONING-HEATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers sheet steel pipe and elbows for air conditioning and heating equipment.

1.2 Classification. Pipe and elbow sections shall be of the following styles and sizes as specified (see 6.2):

Style 1 - Nonadjustable, 90 degree angle
Style 2 - Adjustable, 90 degrees to 0 degrees, and 180 degrees to 90 degrees
Style 3 - Nonadjustable, 45 degree angle
Style 4 - Straight section

Size - 4 inch diameter
Size - 5 inch diameter
Size - 6 inch diameter
Size - 7 inch diameter
Size - 8 inch diameter
Size - 9 inch diameter
Size - 10 inch diameter

Class 1 - Oxidized, dark blue-black color
Class 2 - Zinc coated or hot dipped galvanized

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4520

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2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-20	-	Bags, Cotton, Mailing
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Locked-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-655	-	Boxes, Fiberboard, Six or Eight Sides
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-60	-	Tape, Pressure-Sensitive, Adhesive, Waterproof, for Packaging

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-4-894	-	Pipes and Elbows, Air Conditioning-Heating; Elbows, Nonadjustable, 90 Degrees
5-4-895	-	Pipes and Elbows, Air Conditioning-Heating; Elbows, Adjustable, 90 Degrees
5-4-896	-	Pipes and Elbows, Air Conditioning-Heating; Lengths
5-4-897	-	Pipes and Elbows, Air Conditioning-Heating; Non-adjustable, 45 Degrees

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(Copies of documents required by manufacturers in connection with specific functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality
A 569 - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials. Materials and components shall be as specified herein and in the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all provisions of this document.

3.2.1 Steel sheet, carbon, cold or hot rolled. Cold rolled carbon steel sheet, commercial quality, matte (dull) finish shall conform to ASTM A 366. Hot rolled carbon steel, sheet or strip shall conform to A 569. The basic sheet steel shall be oxidized to produce a uniform, tightly adherent, dark blue-black color or zinc coated in accordance with type 1, SC4 of ASTM B 633 or hot dipped galvanized.

3.3 Construction. The construction of the pipes and elbows shall conform to the requirements specified herein, and as shown on Drawings 5-4-894 through 5-4-897. The class 1 pipes and elbows shall be an oxidized, dark blue-black color. The class 2 pipes and elbows shall be zinc coated or a hot dipped galvanized.

3.3.1 Elbows.

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3.3.1.1 Style 1 and 3 nonadjustable. The nonadjustable elbow shall be formed from one piece of sheet steel specified in 3.2.1 and shall be of corrugated construction. The seaming shall be so finished as to present a uniform inside surface on the plain end and a uniform outside surface at the crimped end. The seams shall not deform these two surfaces.

3.3.1.2 Style 2, adjustable. The adjustable elbow shall be fabricated from not more than four pieces of sheet steel specified in 3.2.1. When in the 90 degree position, the adjustable elbow shall be in accordance with the dimensions specified on Drawing 5-4-895. Elbow seams shall be of the overlapping type secured by not less than two rivets per sheet. The seaming shall be so finished as to prevent a uniform inside surface at the plain end and a uniform outside surface at the crimped end. The elbow shall be manually adjustable from a 90 degree bend to a straight pipe section. The seam for adjustability shall be crimped and overlapping. In addition, the method of crimping shall prevent separation at the adjustable joint.

3.3.2 Pipes. Pipes shall be fabricated from one piece of sheet steel. The longitudinal lock seam shall be of the snap locking design shown on Drawing 5-4-896.

3.4 Instruction sheet. The contractor shall furnish one illustrated instruction sheet for assembly for every five straight pipe sections. An instruction sheet is not required for elbows.

3.5 Marking for identification.

3.5.1 Pipes. Each pipe shall have embossed on its surface the notation "4 inch" (or applicable size) in letters 5/16 inch high followed below by the manufacturer's identification. Letters and figures other than the pipe size shall be not less than 3/16 inch or more than 5/16 inch high. The embossing shall be confined to a space of not more than 4 inches and not less than 1/2 inch from the female or plain end of the pipe.

3.5.2 Elbows. Each elbow shall be marked as specified in 3.5.1 except that the embossing shall be confined to a space of not more than 2 inches and not less than 1/4 inch from the female end of the elbow.

3.6 Workmanship. Rivets shall be driven so that no projection beyond inside surfaces of female and outside surface of male end exists causing any obstruction to assembly of mating ends. The pipes and elbows shall be free of burrs and slivers. The fabricated steel sheet finish shall be free of rust or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or

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purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of pipes or elbows of one style and size only. The sample unit shall be one pipe or elbow. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not a uniform dark blue black oxide color or not zinc coated, as applicable	X	
	Rust or foreign matter		X
Design, construction and workmanship	Any malformation, damage, fracture or dents	X	
	Sharp edges, burrs, or sliver	X	
	Rivet on elbow cocked or head protrudes inside female end or outside male end		X
	Adjustable elbows not workable manually without deformation	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Marking, identification and instruction sheet	Missing, incomplete, or not legible; size incorrect, not in proper location		X

4.4.3 End item dimansional examination. The end item shall be examined for conformance to the dimensions specified in this document and on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of pipes or elbows. The sample unit shall be one pipe or elbow of one style and size only. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully packaged, except it need not be sealed. The lot size shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE II. Packaging defects

Examine	Defect
Marking (interior pack or container and exterior container as applicable)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservative application	Improperly applied or missing
Materials	Component missing, damaged, defective, or not as specified
Contents (interior and exterior container as applicable)	Number per container not as specified
Content	More or less than required

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative application.

5.1.1.1.1 Pipe sections and elbows. Each elbow or pipe section except when zinc coated, or hot dip galvanized, shall be cleaned process C-1, thoroughly dried, and coated with type P-7 preservative in accordance with MIL-P-116. When elbows and pipe joints are made from steel, which is treated in the continuous bluing process and is heated to a sufficiently high temperature so as to burn all surfaces clean and free from grease and dirt, and in which process it is cooled, and coated with type P-7 preservative, the cleaning process specified above shall be omitted. Steel must be handled with gloves, and precaution taken to avoid contact of steel surfaces with possible contaminants.

5.1.1.1.2 Instruction sheets. The instruction sheets shall be preserved method IC-1 of MIL-P-116. Instruction sheets used with zinc-coated pipe or hot dip galvanized sections may be placed in a kraft paper envelope in lieu of method IC-1 specified herein.

5.1.1.2 Unit packing.

5.1.1.2.1 Elbows. Four style 1 or 2 elbows, or eight style 3 elbows of one size and finish only, preserved as specified in 5.1.1.1.1, shall be interlocked together to form a hollow square or octagon, respectively. Each group of elbows shall then be wrapped in greaseproof barrier material conforming to type I or II, grade A of MIL-B-121, and the wrap secured with jute, sisal, manila, or cotton cord, or twine, or tape conforming to PPP-T-60 or PPP-T-45, or shall be heat sealed when heat sealable barrier material is used. Zinc-coated elbows shall not require the barrier wrap.

5.1.1.2.2 Pipe sections. Twenty-five sections of pipe (size 4-inch and 5-inch) or ten sections of pipe (sizes 6-inch through 10-inch inclusive) of one size and finish only, preserved as specified in 5.1.1.1.1 shall be nested to form a cylinder. The rivets shall be placed in a cloth drawstring bag conforming to PPP-B-20. The bag shall be tightly drawn and tied to prevent loss of contents. The bagged rivets along with the instruction sheets, preserved as specified in 5.1.1.1.2, shall be rolled to the proper diameter and placed within the void of the nested pipe sections. The nested pipe sections shall then be completely wrapped in barrier material, as specified in 5.1.1.2.1. Zinc-coated pipe sections shall not require the barrier wrap. The pipe sections shall then be placed in a snug-fitting fiberboard box conforming to type I, style C of PPP-B-655. When applicable, fiberboard boxes conforming to style RSC (end opening), type CF, variety optional, class domestic of PPP-B-636, may be used. closure shall be in accordance with the appendix of the applicable box specification.

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5.1.1.3 Intermediate packing. Twelve style 1 or 2 elbows or twenty-four style 3 elbows of one size and finish only, unit packed as specified in 5.1.1.2.1, shall be packed in a snug-fitting intermediate fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636 or type II, style C of PPP-B-655. Unit packs shall be placed flat, three in depth within the box. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Commercial. Pipe sections, elbows, rivets and instruction sheets shall be preserved and packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Elbows. Forty-eight elbows of one style, size and finish only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to grade V2s, style RSC of PPP-B-636; overseas type of PPP-B-601 or class 2, style 2 or 4 of PPP-B-621. Each fiberboard shipping container shall be closed and waterproofed by means of tape and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636. Shipping containers other than fiberboard shall be closed and strapped in accordance with the appendix of the applicable container specification. The weight of contents for fiberboard shipping containers shall be subject to the weight limitations of PPP-B-636.

5.2.1.2 Pipe sections. Fifty sections of pipe (sizes 4-inch and 5-inch) or thirty sections of pipe (sizes 6-inch through 10-inch inclusive) of one size and finish only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type of PPP-B-601 or class 2, style 2 or 4 of PPP-B-621. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification.

5.2.2 Level B packing.

5.2.2.1 Elbow. Twelve style 1 or 2 elbows, or twenty-four style 3 elbows of one size and finish only, preserved as specified in 5.1.1, shall require no additional packing.

5.2.2.2 Pipe sections. Twenty-five sections of pipe (sizes 4-inch and 5-inch) or ten sections of pipe (size 6-inch through 10-inch inclusive) of one size and finish only, preserved as specified in 5.1.1 shall require no additional packing.

5.2.3 Commercial packing. Elbows and pipe sections, preserved as specified in 5.1, shall be packed separately in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The pipe and elbows are intended for use with Military Air Conditioning, Heating and Cooking Equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Style, size and class of pipe and elbow required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of the applicable levels of preservation, and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one item. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-551 E	Pipes And Elbows, Air Conditioning-Heating

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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INCH-POUND

MIL-P-551E
AMENDMENT-2
21 November 1988
SUPERSEDING
AMENDMENT-1
17 April 1986

MILITARY SPECIFICATION

PIPES AND ELBOWS, AIR CONDITIONING-HEATING

This amendment forms a part of Military Specification MIL-P-551E, dated 21 September 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

* 3.3.2, second sentence: Delete and substitute "The longitudinal lock seam design shall be a commercial Reeves or Modified Reeves type."

3.5.1, line 3, after "identification": Add "and year of manufacture."

PAGE 6

Add the following new paragraphs:

* "4.4.5 End item testing. Testing of the completely finished pipe section and smokestack assembly shall be performed in accordance with 4.4.5.1. Any test failure shall be counted as a defect. The lot size shall be expressed in units of completely finished pipes or elbows, as applicable, of one style, size, and class. The sample unit shall be one elbow or pipe section. The inspection level shall be S-4 and the AQL, expressed in defects per hundred units, shall be 1.0."

* "4.4.5.1 Assembly test. Using the instruction sheet provided by contractor, as required in 3.4, a pipe section shall be assembled. Inability to interlock the locking seam along the entire length of the pipe shall constitute a test failure. After satisfactorily assembling the pipe section, three pipe sections shall be mated together firmly enough to seat one pipe against the other. This will simulate field assembly of a smoke stack. Locking seams of all three pipes shall remain interlocked along entire length of pipe. Inability of the pipe sections to mate together or the locking seam to remain interlocked along the entire length of pipe, shall constitute a test failure."

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* 5.1.1.2.2, second, third, fourth sentences: Delete and substitute "The instruction sheets, preserved as specified in 5.1.1.1.2, shall be rolled to proper diameter and placed within the void of the nested pipe sections."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and cotractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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